

Ritika

Senior Developer

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Summary

Senior Developer experienced in building industrial web applications, backend APIs, database workflows, and AI-powered solutions. Combines software development expertise in **Python, FastAPI, SQL Server, and JavaScript** with hands-on experience in **RAG, speech AI, large-scale data pipelines, and ML model evaluation**.

Experience

Senior Developer

Key Point Technologies

Client: Chennai Petroleum Corporation Limited (CPCL)

Jul 2026 – Present

Chennai, India | On-site

- Developing an industrial **Heat Exchanger monitoring and calculation application** using **Python, FastAPI, SQL Server, HTML, CSS, and JavaScript**, integrating live process data, engineering calculations, and historical records.
- Implementing **backend APIs, database workflows, and frontend views** for current calculations and history; preparing FastAPI deployment on **IIS**, including dependencies and `web.config` planning.
- Contributing to a **RAG-based Safety Manual Assistant** through PDF inspection, chapter mapping, and retrieval-pipeline and local-LLM planning under enterprise API and security constraints.

Project Associate

SPIRE Lab, Indian Institute of Science (IISc)

Client: Google

Oct 2023 – Mar 2026

Bengaluru, India | On-site

- Fine-tuned **Wav2Vec2-based ASR models** on the **Respin** dataset using **Fairseq**, leveraging LoRA and bottleneck adapters to improve performance.
- Built training and evaluation pipelines using **WER/CER** and automated quality analysis across multiple **Indic speech datasets** for code-switching and speech naturalness.
- Analyzed **70 lakh+ speech records** using Python/Bash automation for cleaning and demographic insights; supported **Project Vaani** image validation with FRCNN/MTCNN-based quality checks.

Student Trainee

Nokia Corporation

Aug 2022 – Jul 2023

Bengaluru, India | Remote

- Supported development, testing, and debugging for **Alarm Generator** and **Unified Inventory** telecom projects on Red Hat Enterprise Linux using Git and Jira.

Projects

Discharge AI Agent

May 2026

Built a clinical summarization agent with custom workflows, safety guardrails, conflict detection, and iterative correction.

Airline Customer Support RAG Assistant

Jun 2026

Built a Streamlit RAG assistant with three evaluated chunking strategies, retrieval benchmarks (Precision@K, Recall@K, MRR), and grounding guardrails.

Publications

VAANI: Capturing the Language Landscape for an Inclusive Digital India

2026

[arXiv:2603.28714](https://arxiv.org/abs/2603.28714) (Submitted to NeurIPS 2026, Evaluations and Datasets Track)

Education

B.Tech in Computer Science and Engineering

Galgotias University

2019 – 2023

Skills

Programming & Backend: Python, SQL, FastAPI, REST APIs, SQL Server, HTML, CSS, JavaScript

Machine Learning & GenAI: PyTorch, Hugging Face, LLMs, RAG, AI Agents, Prompt Engineering

Speech & Evaluation: ASR, Wav2Vec2, WhisperX, Fairseq, LoRA, Adapters, WER/CER/DER, Dataset Quality

Tools & Systems: Git, Bash, Linux, IIS, OpenCV, WandB, FRCNN, MTCNN